

Governance Effects and the Role of On and Off-Chain Vote Divergence in Decentralized Autonomous Organizations (DAOs)

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1. Introduction

Governance effects play a very important role in financial markets, with prior studies proving that passing governance proposals generates positive value both in corporate settings and in decentralized autonomous organizations (DAOs). However, DAO governance is facing a unique institutional feature, which is systematic divergence between on-chain and off-chain voting outcomes. This study uses DAO platform data and a regression discontinuity design to show that while positive governance effects persist, vote divergence weakens governance value by attenuating the effectiveness of governance decisions.

2. Literature Review and a New Challenge in DAOs

A central question in the corporate governance literature concerns the causal effect of governance decisions on firm value. Using close-call shareholder proposals in U.S. public firms, Cuñat et al. (2012) show that passing governance-related proposals generates significant positive abnormal stock returns, establishing a causal link between governance and firm value. This identification framework has recently been extended to decentralized autonomous organizations (DAOs): Appel and Grennan (2024) apply a regression discontinuity design to DAO governance proposals and find that passing inclusive or security-enhancing proposals increases token returns and improves real outcomes such as trading volume and user growth.

Despite these documented governance effects, DAO governance features a distinctive institutional structure—the coexistence of on-chain and off-chain voting. On-chain voting is costly and dominated by large token holders, whereas off-chain voting attracts broader community participation. As shown in Figure 1, this asymmetry in participation is accompanied by systematic outcome divergence: within our sample, on-chain and off-chain yes-share often differ substantially, and approximately 16% of proposals yield conflicting pass/fail outcomes (for more divergence analysis, see [Dune Dashboard](#)). These patterns motivate our central question: does divergence between on-chain and off-chain voting outcomes weaken the value that governance creates?

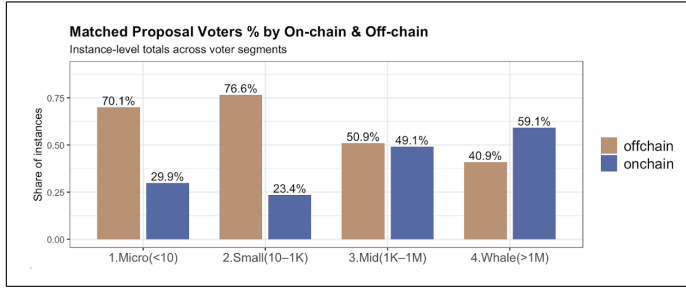


Figure 1. Participation Levels in On-Chain and Off-Chain Voting by User Segment

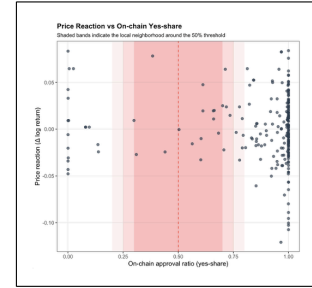


Figure 2. RDD around the 50% On-Chain Approval Threshold

3. Data and Empirical Strategy

Our sample consists of governance proposals from January 2021 to November 2025 across four major DAOs—Uniswap, Arbitrum, Gitcoin, and Optimism—collected from the Dune platform (for data extract details, see [Dune Query](#)). On-chain and off-chain records for the same proposals are linked using semantic matching and manually verified, resulting in a final sample of 229 proposals.

Following prior studies, our primary outcome is cryptocurrency log returns measured in a short event window around proposal outcomes, capturing immediate market reactions to governance decisions (Cuñat et al., 2012; Appel and Grennan, 2024). The key explanatory variables include an indicator for whether a proposal passes and its interaction with the on–off voting gap, which measures divergence in yes-share between the two systems. Our main hypotheses are:

H1: Passing a governance proposal generates positive governance value.

H2: Greater vote divergence weakens the governance value created by passing proposals.

To identify causal governance effects, we employ a regression discontinuity design that exploits the 50% approval threshold as a quasi-experiment as shown in Figure 2. This approach mitigates omitted variable concerns, as DAOs with stronger governance may also differ systematically in community engagement, developer quality, or market positioning. By comparing proposals that narrowly pass or fail, the RDD framework isolates the causal effect of passing proposals and follows established practice in both corporate and DAO governance research (Lee, 2008; Park et al., 2015; Cuñat et al., 2012; Appel and Grennan, 2024).

4. Results

As shown in Table 1, the results provide evidence for two main findings. First, governance decisions in DAOs generate positive market reactions: the coefficient on Pass is positive across local RDD specifications, indicating that proposals that narrowly pass tend to be associated with higher returns. Second, vote divergence between on-chain and off-chain outcomes significantly weakens this positive governance effect. The interaction term **Pass × gap** is consistently negative and statistically significant in the local bandwidth specifications. For example, the estimate of -0.658 in model (1) implies that a 0.01 increase in the on–off yes-share gap reduces the governance effect of passing a proposal by

approximately 0.66 percentage points, suggesting that divergence attenuates the value created by governance decisions.

These findings are robust across alternative bandwidth choices around the 50% approval threshold. The patterns are consistent across bandwidths of 0.20, 0.25, and 0.30, particularly with respect to the negative moderating role of vote divergence. While not all coefficients remain statistically significant when the full sample is used without bandwidth restrictions, this is expected given the loss of local quasi-random variation and the relatively small number of observations near the cutoff. Overall, the results support the robustness of the main conclusions within the local neighborhood where the RDD identification is most credible. In addition, to assess outcome robustness, we estimate log returns over alternative event windows ranging from 2 to 10 days; the coefficients on key variables remain qualitatively similar in sign across specifications, as shown in Figure 3.

	(1) h = 0.20	(2) h = 0.25	(3) h = 0.30	(4) All
Constant	-0.057 (0.038)	-0.055 (0.038)	-0.084* (0.040)	0.011 (0.018)
Pass	0.113 (0.059)	0.119* (0.057)	0.147** (0.057)	0.022 (0.028)
Governance gap	0.608** (0.165)	0.603*** (0.189)	0.663*** (0.214)	0.004 (0.039)
Pass × gap	-0.658*** (0.177)	-0.687*** (0.195)	-0.732*** (0.218)	-0.031 (0.043)
Off-chain pass	-0.303*** (0.067)	-0.302*** (0.079)	-0.314*** (0.090)	-0.028 (0.028)
Pass × off_pass	0.249** (0.070)	0.246** (0.082)	0.274*** (0.092)	0.009 (0.031)
Running var.	0.081 (0.120)	0.091 (0.090)	-0.041 (0.071)	-0.025 (0.023)
R-squared	0.837	0.696	0.485	0.026
N	13	18	24	229

* h denotes the bandwidth ($(\text{running_var} \leq h)$; smaller h restricts the sample closer to the 50% approval threshold.
† 'All' uses the entire sample without any bandwidth restriction.

Table 1. RDD Estimates Result

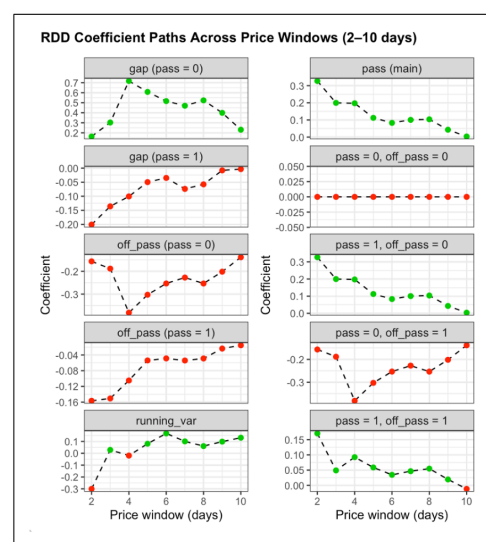


Figure 3. RDD Coefficient Stability across Price Windows

5. Conclusion and Discussion

This study provides evidence that governance decisions in DAOs generate positive market value, consistent with prior findings in both corporate and decentralized governance settings. At the same time, we show that divergence between on-chain and off-chain voting outcomes weakens governance value by attenuating the effectiveness of passing proposals. This negative moderating effect suggests that off-chain voting captures valuable information about broader community preferences that is not fully reflected in on-chain outcomes. Taken together, the results highlight that governance quality in DAOs depends not only on whether proposals pass, but also on how aligned different voting mechanisms are in aggregating preferences.

Several avenues remain for future research. First, data limitations constrain the current analysis to a relatively small sample of proposals from a single platform; future work could expand the sample by incorporating additional DAOs, governance platforms, and longer time horizons.

Second, while this study focuses on short-term price reactions measured by log returns, future research could examine richer outcome measures—such as trading volume, user activity, protocol usage, or other real outcomes—following the approach of Appel and Grennan (2024), and assess whether governance effects persist in the longer run. Beyond data expansion, future work could further explore the drivers and consequences of on–off voting divergence, including vote concentration, delegation, participation costs, and timing frictions, as well as its implications for legitimacy and market responses.

Finally, drawing parallels to strategic voting in political elections, future research could investigate whether off-chain voting signals induce strategic behavior—such as deserting weaker proposals in favor of perceived frontrunners—in more complex, multi-proposal governance environments (Lijphart, 1997; Lee, 2008).

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